INTRODUCTION TO CLINICAL TRIALS & CLINICAL DATA MANAGEMENT



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09 Jan 2026

MARKS DISTRIBUTION

- Total Marks 100
- Every class a quiz which will be evaluated - 25 marks
- Project – 30 marks
- End semester exam – 45 marks.

OBJECTIVES

- Understand the basics of clinical trials
- Learn the phases and design of trials
- Importance of Good Clinical Practice (GCP)
- Role of Clinical Data Management (CDM)
- Overview of CDM process

WHAT ARE CLINICAL TRIALS ?

- Research studies in humans to evaluate safety & efficacy of drugs, devices, or interventions
- Bridge between laboratory research → patient care
- Essential for regulatory approval

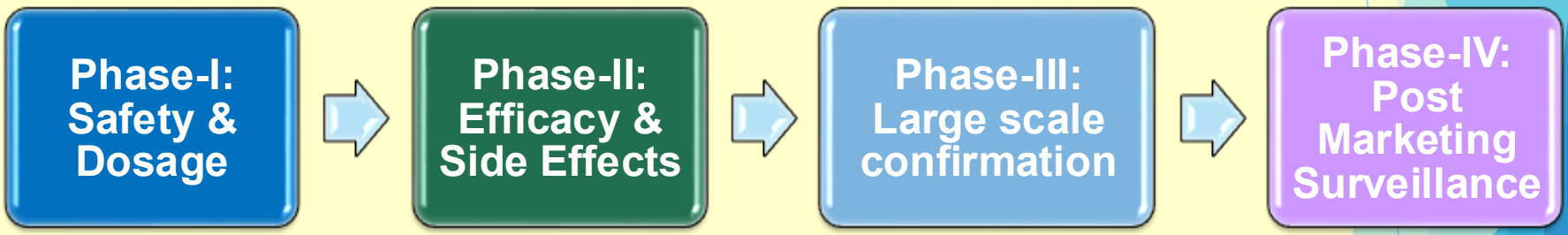
IMPORTANCE OF CLINICAL TRIALS

- Ensure patient safety
- Determine drug effectiveness
- Provide evidence for regulatory approval
- Advance medical knowledge

PHASES OF CLINICAL TRIALS

- Phase I – Safety & dosage (20–100 subjects)
- Phase II – Efficacy & side effects (100–300 subjects)
- Phase III – Large-scale confirmation (1,000–3,000 subjects)
- Phase IV – Post-marketing surveillance

FLOWCHART: PHASES OF CLINICAL TRIALS



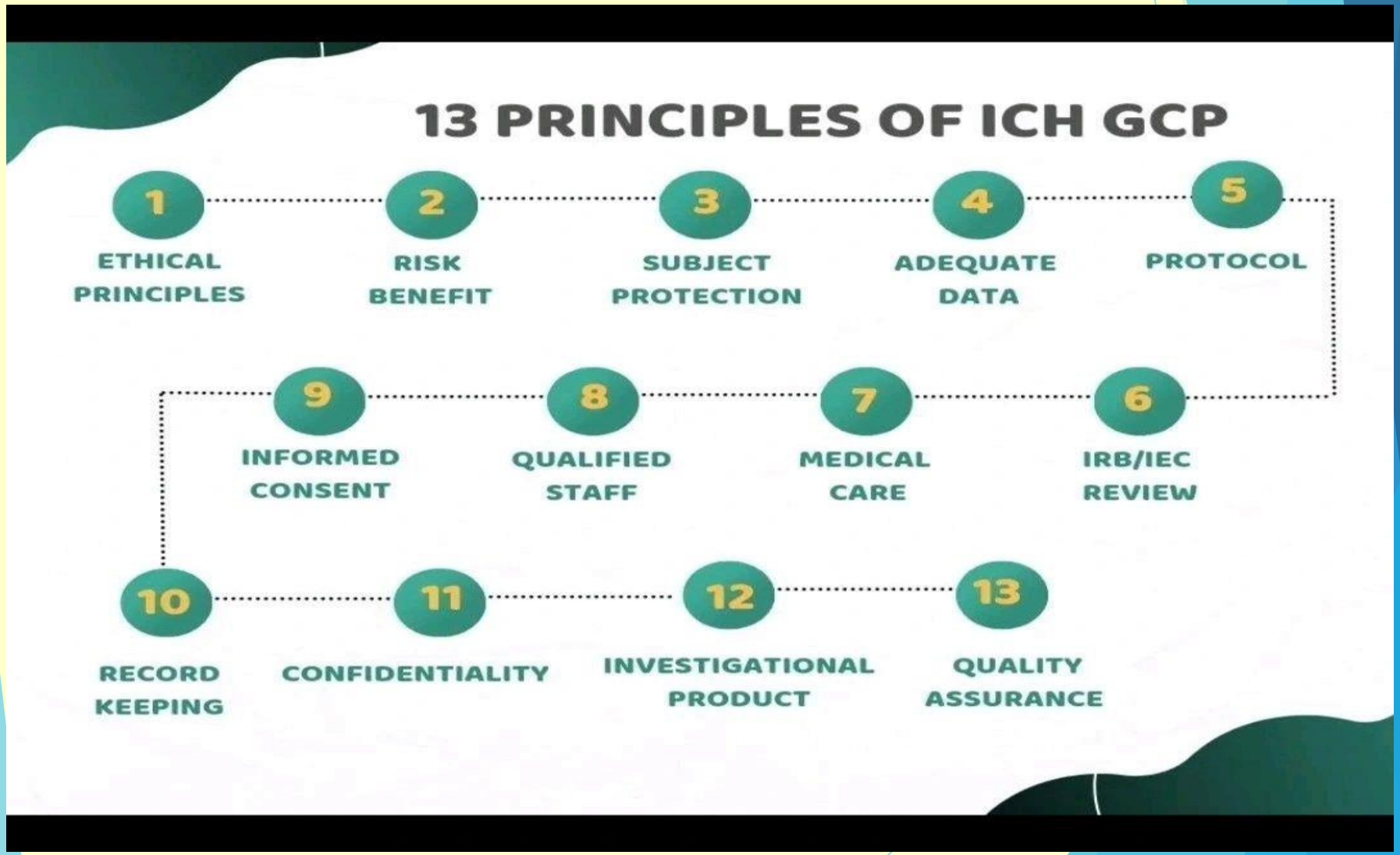
DESIGN OF CLINICAL TRIALS

- Randomized Controlled Trial (RCT)
- Double-Blind Study
- Placebo-Controlled Trial
- Crossover & Observational Studies

GOOD CLINICAL PRACTICE (GCP)

- Ethical & scientific quality standard
- Protects human rights, safety, and well-being
- Ensures credibility of data
- Governed by International Council of Harmonisation (ICH)-Good Clinical Practice (GCP) guidelines

PRINCIPLES OF ICH GCP



CONCLUSION

- Clinical trials → backbone of medical advancement
- CDM → ensures data integrity and regulatory compliance
- Both essential for safe & effective healthcare innovation

REFERENCES

- ICH-GCP Guidelines
- FDA Clinical Trial Regulations
- Textbooks on Clinical Research & Data Management